

Q1 - 25 July - Shift 1

Which of the following statement is correct ?

Space for your notes:

(A) In primary rainbow, observer sees red colour on the top and violet on the bottom

(B) In primary rainbow, observer sees violet colour on the top and red on the bottom

(C) In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops

(D) Primary rainbow is less bright than secondary rainbow.

Q2 - 25 July - Shift 1

Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B

of same thickness is t_1 and t_2 respectively. If

$t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is 1 : 2.

Then the thickness of material, in meter is : (Given

v_A and v_B are velocities of light in A and B

materials respectively).

(A) $5 \times 10^{-10} v_A$ m

(B) 5×10^{-10} m

(C) 1.5×10^{-10} m

(D) $5 \times 10^{-10} v_B$ m

Q3 - 25 July - Shift 2

For an object placed at a distance 2.4 m from a lens, a sharp focused image is observed on a screen placed at a distance 12 cm from the lens. A glass plate of refractive index 1.5 and thickness 1 cm is introduced between lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen?

- (A) 0.8 m (B) 3.2 m
(C) 1.2 m (D) 5.6 m

Space for your notes:

Q4 - 25 July - Shift 2

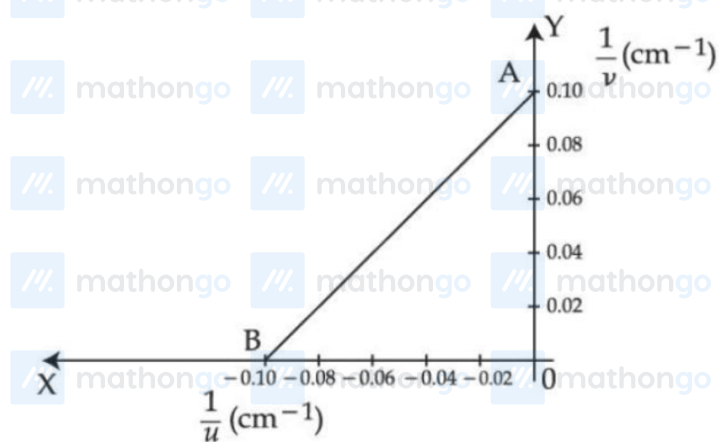
A convex lens of focal length 20 cm is placed in front of convex mirror with principal axis coinciding each other. The distance between the lens and mirror is 10 cm. A point object is placed on principal axis at a distance of 60 cm from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is _____ cm.

Space for your notes:

Q5 - 26 July - Shift 1

The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvatures R . The value of R will be _____ cm.

(Where u = object distance, v = image distance)



Space for your notes:

Q6 - 26 July - Shift 2

Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8 \text{ ms}^{-1}$ and $2.0 \times 10^8 \text{ ms}^{-1}$ respectively. The critical angle between them is:

- (A) $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$
 (C) $\cos^{-1}\left(\frac{3}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

Space for your notes:

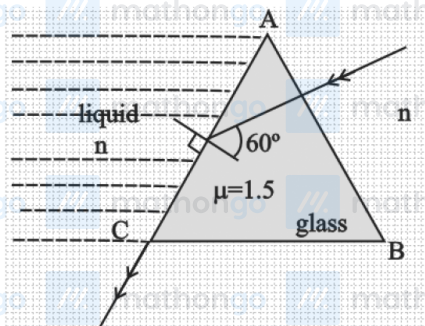
Q7 - 26 July - Shift 2

#MathBoleTohMathonGo

In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC.

The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The

value of x is _____.
(Given refractive index of glass = 1.5)



Space for your notes:

Q8 - 27 July - Shift 1

A microscope was initially placed in air (refractive index 1). It is then immersed in oil (refractive index 2). For a light whose wavelength in air is λ , calculate the change of microscope's resolving power due to oil and choose the correct option

(A) Resolving power will be $\frac{1}{4}$ in the oil than it

was in the air

(B) Resolving power will be twice in the oil than it was in the air.

(C) Resolving power will be four times in the oil than it was in the air.

(D) Resolving power will be $\frac{1}{2}$ in the oil than it was in the air.

Space for your notes:

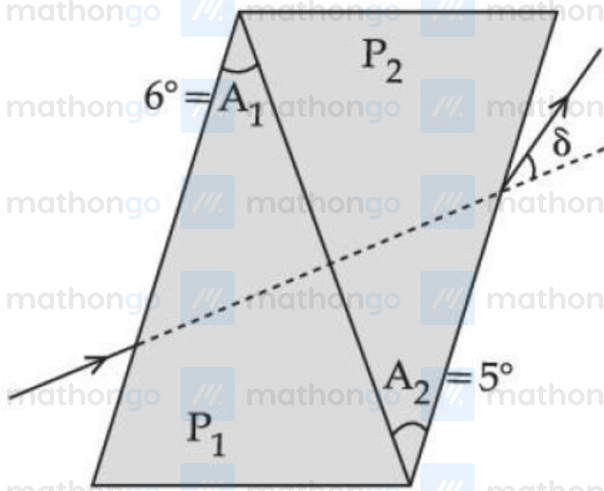
Q9 - 27 July - Shift 2

Questions

MathonGo

A thin prism of angle 6° and refractive index for yellow light (n_Y) 1.5 is combined with another prism of angle 5° and $n_Y = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is.....

Space for your notes:

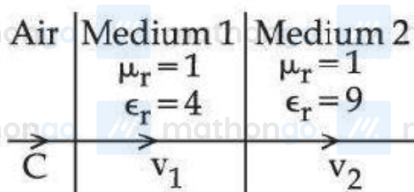


Q10 - 28 July - Shift 1

As shown in the figure, after passing through the medium 1. The speed of light v_2 in medium 2 will be :

Space for your notes:

(Given $c = 3 \times 10^8 \text{ ms}^{-1}$)



(A) $1.0 \times 10^8 \text{ ms}^{-1}$ (B) $0.5 \times 10^8 \text{ ms}^{-1}$

(C) $1.5 \times 10^8 \text{ ms}^{-1}$ (D) $3.0 \times 10^8 \text{ ms}^{-1}$

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Q11 - 28 July - Shift 1

In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be:

- (A) 20 cm (B) 30 cm
(C) 10 cm (D) 15 cm

Space for your notes:

Q12 - 28 July - Shift 2

The power of a lens (biconvex) is 1.25 m^{-1} in particular medium. Refractive index of the lens is 1.5 and radii of curvature are 20 cm and 40 cm respectively. The refractive index of surrounding medium :

- (A) 1.0 (B) $\frac{9}{7}$
(C) $\frac{3}{2}$ (D) $\frac{4}{3}$

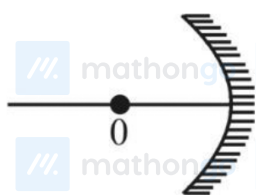
Space for your notes:

Q13 - 28 July - Shift 2

Questions

MathonGo

An object 'o' is placed at a distance of 100 cm in front of a concave mirror of radius of curvature 200 cm as shown in the figure. The object starts moving towards the mirror at a speed 2 cm/s. The position of the image from the mirror after 10s will be at cm.

*Space for your notes:***Q14 - 29 July - Shift 1**

The X-Y plane be taken as the boundary between two transparent media M_1 and M_2 . M_1 in $Z \geq 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{A} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be _____ degree.

*Space for your notes:***Q15 - 29 July - Shift 2**

#MathBoleTohMathonGo

Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of 15° from its original direction.

The refractive index of the medium is :

- (A) 1.732 (B) 1.333
(C) 1.414 (D) 2.732

Space for your notes:

Answer Key

Q1 (A)

Q2 (A)

Q3 (B)

Q4 (10)

Q5 (10)

Q6 (A)

Q7 (27)

Q8 (B)

Q9 (4)

Q10 (A)

Q11 (A)

Q12 (D)

Q13 (400)

Q14 (15)

Q15 (C)

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Q1 (A)

In primary rainbow, red colour is at top and violet is at bottom.

Intensity of secondary rainbow is less in comparison to primary rainbow.

Q2 (A)

$$\frac{\mu_A}{\mu_B} = \frac{c/V_A}{c/V_B} = \frac{V_B}{V_A} = \frac{1}{2}$$

Let the thickness is d

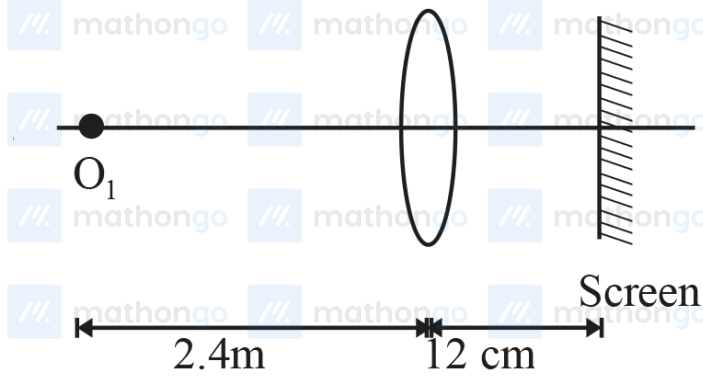
$$\frac{d}{V_B} - \frac{d}{V_A} = 5 \times 10^{-10}$$

$$d = \frac{5 \times 10^{-10} \times V_A V_B}{V_A - V_B}$$

$$\text{As } v_A = 2v_B \Rightarrow d = 5 \times 10^{-10} \times 2v_B$$

$$\text{Or } d = 5 \times 10^{-10} \times v_A$$

Q3 (B)



Applying lens formula

$$\frac{1}{0.12} + \frac{1}{2.4} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{210}{24}$$

Upon putting the glass slab, shift of image is

$$\Delta x = t \left(1 - \frac{1}{\mu} \right) = \frac{1}{3} \text{ cm}$$

$$\text{Now } v = 12 - \frac{1}{3} = \frac{35}{3} \text{ cm}$$

Again apply lens formula

$$\frac{1}{0.12} + \frac{1}{u} = \frac{1}{f} = \frac{210}{24}$$

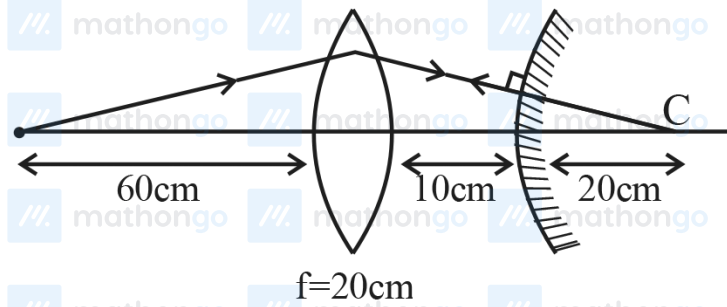
Solving $u = -5.6 \text{ m}$

Thus shift of object is

$$5.6 - 2.4 = 3.2 \text{ m}$$

Q4 (10)

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For lens

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{(-60)} = \frac{1}{20} \Rightarrow \frac{1}{v} + \frac{1}{60} = \frac{1}{20}$$

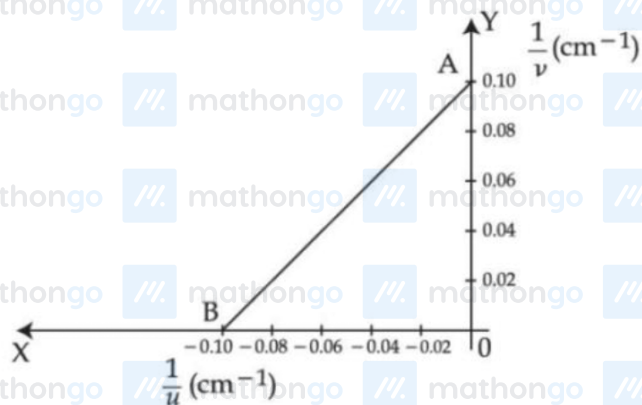
$$v = 30 \text{ cm}$$

For final image to be formed on the object itself, after refraction from lens the ray should meet the mirror perpendicularly and the image by lens should be on the centre of curvature of mirror

$$R = 30 - 10 = 20 \text{ cm}$$

$$\text{Focal length of mirror} = R/2 = 10 \text{ cm}$$

Q5 (10)



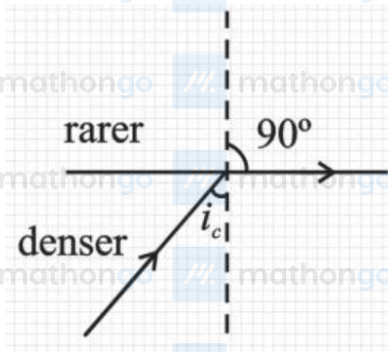
For point B, $\frac{1}{u} = -0.10 \text{ cm}^{-1}$, $\frac{1}{v} = 0$

$\therefore$ Thus, $u = -10 \text{ cm}$, $v = \infty$

i.e. $f = 10 \text{ cm}$

$$\Rightarrow \frac{1}{10 \text{ cm}} = (1.5 - 1) \left(\frac{2}{R} \right) = \frac{1}{R} \Rightarrow R = 10 \text{ cm}$$

Q6 (A)



$$v = \frac{c}{n}$$

$$n_d \sin i_c = n_r \sin 90^\circ$$

$$\sin i_c = \frac{n_r}{n_d} = \frac{v_d}{v_r}$$

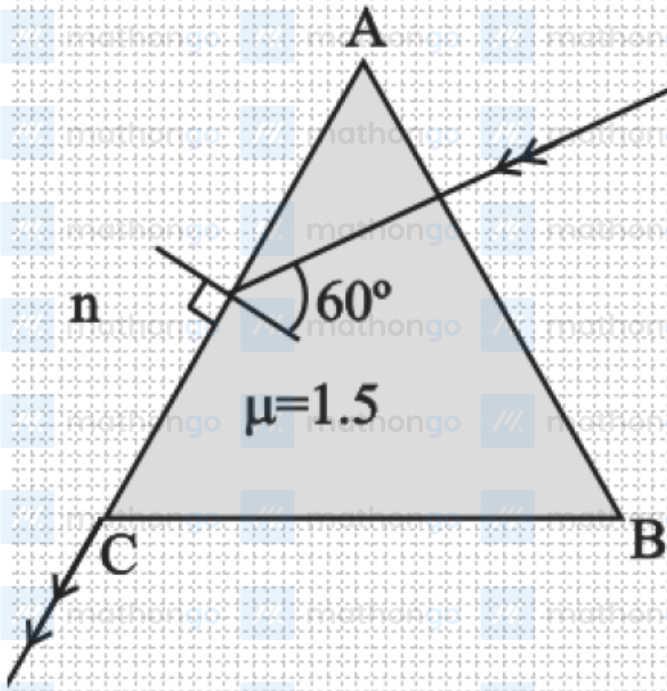
$$\sin i_c = \frac{1.5 \times 10^8}{2 \times 10^8} = \frac{1.5}{2}$$

$$\sin i_c = \frac{3}{4}$$

$$\tan i_c = \frac{3}{\sqrt{4^2 - 3^2}} \Rightarrow \frac{3}{\sqrt{7}}$$

$$i_c = \tan^{-1} \left(\frac{3}{\sqrt{7}} \right)$$

Q7 (27)



Using snell's law at face AC

$$1.5 \sin 60^\circ = n \times \sin 90^\circ$$

$$1.5 \times \frac{\sqrt{3}}{2} = n = \frac{\sqrt{x}}{4}$$

$$3\sqrt{3} = \sqrt{x}$$

$$x = 27$$

Q8 (B)

Q9 (A)

$$(R.P)_{\text{air}} = \frac{2 \sin \theta}{1.22 \lambda}$$

$$(R.P)_{\text{oil}} = \frac{2 \sin \theta}{1.22 \lambda_{\text{oil}}} = \frac{2 \sin \theta \times \mu}{1.22 \lambda}$$

$$(R.P)_{\text{oil}} = (R.P)_{\text{air}} \times 2$$

Q9 (4)

$$\delta = A(\mu_y - 1) - A'(\mu_{y'} - 1)$$

$$= 6(1.5 - 1) - 5(1.55 - 1)$$

$$= \frac{1}{4}$$

Q10 (A)

$$\underline{\mu_2} = \underline{\frac{C}{v_2}}$$
$$\mu_{\text{air}} = \frac{C}{v_2}$$

$$\therefore \frac{\sqrt{\mu_{r_2} \epsilon_{r_2}}}{(1)} = \frac{C}{v_2}$$

$$\therefore \sqrt{(1)(9)} = \frac{C}{v_2}$$

$$\therefore v_2 = \frac{C}{3}$$

Q11 (A)

$$f_o + f_e = 30$$

$$m = \frac{f_o}{f_e}$$

$$2 = \frac{f_o}{f_e} \Rightarrow f_o = 2f_e$$

$$\text{So } f_o + \frac{f_o}{2} = 30$$

$$f_o = 20 \text{ cm}$$

Q12 (D)

$$P = \frac{\mu_2}{f} = (\mu_1 - \mu_2) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (\text{For this formula}$$

refer to NCERT Part-2, Chapter-9, Page no. 328,

solved example 8)

(μ_1 is refractive index of lens and μ_2 is of surrounding medium)

$$1.25 = (1.5 - \mu_2) \left(\frac{1}{0.2} + \frac{1}{0.4} \right)$$

$$\frac{1.25 \times 0.08}{0.6} = (1.5 - \mu_2)$$

$$\Rightarrow \mu_2 = \frac{4}{3}$$

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Q13 (400)

After 10 sec.

$$u = -80 \text{ cm}$$

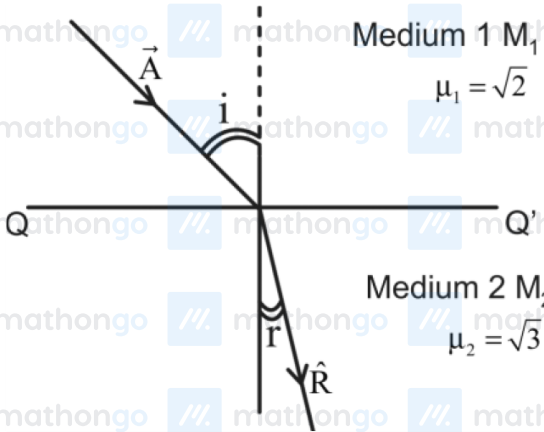
$$f = -100 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$v = 400 \text{ cm}$$

Q14 (15)

$$\vec{A} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$$



$$\mu_1 \sin i = \mu_2 \sin r$$

As incident vector A makes i angle with normal z -axis & refracted vector R makes r angle with normal z -axis with help of direction cosine

$$i = \cos^{-1}\left(\frac{A_z}{A}\right) = \cos^{-1}\left(\frac{5}{\sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2 + 5^2}}\right)$$

$$= \cos^{-1}\left(\frac{5}{10}\right) \Rightarrow i = 60^\circ$$

$$\sqrt{2} \sin 60 = \sqrt{3} \times \sin r$$

$$r = 45^\circ$$

Difference between i & $r = 60 - 45 = 15$

Q15 (C)



$$i = 45^\circ$$

$$D = i - r$$

$$15^\circ = 45^\circ - r \Rightarrow r = 30^\circ$$

$$n_1 \sin i = n_2 \sin r$$

$$1 \sin 45^\circ = \mu \sin 30^\circ$$

$$\frac{1}{\sqrt{2}} = \mu \frac{1}{2}$$

$$\mu = \sqrt{2} = 1.414$$